



Assessment for Frontend Position

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Vrit Technologies
Sankhamul
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In partial fulfillment of the requirements for the Frontend position

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ABSTRACT

The current paper outlines the development process for an interactive React Form Builder, where users can dynamically create forms of their liking with a drag-and-drop functionality. As such, the final project focuses mainly on component composition, form validation, and properly applied state management regarding the identified functional requirements for implementing necessary excerpts and inclusions of bonus features.

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Chapter-1

1.1 INTRODUCTION

A form builder is an application or tool designed to help users create, manage, and customize forms without much technical knowledge or programming skills. This application will make data collection easy because users are allowed to design and then publish forms that will be used for different purposes: surveys, feedback, registration, and order taking.

Key Features of a Form Builder:

1. **Drag-and-Drop Interface:** Allows users to add fields like text input, dropdowns, checkboxes, and radio buttons easily.
2. **Field Customization:** Options to label fields, set placeholder text, and define validation rules (e.g., required fields, email validation).
3. **Responsive Design:** Ensures forms work seamlessly on all devices, including desktops, tablets, and smartphones.
4. **Conditional Logic:** Displays or hides fields based on user responses for a personalized experience.
5. **Styling and Branding:** Options to customize colors, fonts, and layouts to match the organization's brand identity.
6. **Data Storage and Export:** Captures submissions and allows exporting data in formats like JSON format.

Advantages:

- Saves time and effort compared to coding forms manually.
- Reduces dependency on developers for basic form creation.
- Enhances user experience with interactive and visually appealing forms.
- Facilitates efficient data collection and analysis

1.2 OBJECTIVES

- Provide a simple interface, often drag-and-drop, to create forms without requiring coding expertise.
- Allow users to customize form fields, layouts, and design elements effortlessly.
- Reduce the time and effort needed to create and deploy forms.

CHAPTER-2

METHODOLOGY

2.1 Requirement Collection

Requirement analysis for the proposed Form Builder was carried out by considering previous research and existing systems. Information was gathered by analyzing relevant documents and manuals. To the end, the system was integrated. Once the objective was defined, requirement gathering started to gather details on form-building tools, its history, and past initiatives in this area.

The requirement gathering techniques used are as follows:

- **Literary Analysis** Literature related to form-building tools was reviewed and critically analyzed to understand the solutions suggested in the past. This analysis helped identify the shortcomings of existing systems and refine the aim and objectives of the project.
- **Observation:** Several form-building tools were reviewed to identify current trends in developing form-building applications. Popular tools like Google Forms, Typeform, JotForm, and WPForms were analyzed to gain insight into their features, usability, and form submission processes.

2.2 Tools Used While building the Project

To deploy the application, the only technical aspects needed are mentioned below:

- **Hardware Requirement**
Processor: 800MHz Intel Pentium III or equivalent or new
Disk space: 100GB or more
RAM: 128MB or more
- **Software Requirement**
Operating System: Windows (7 or more)
Web Browser: IE 10 or above, Mozilla FF and above or Google Chrome
- **Programming Language:** HTML, CSS, Js, React, Typescript, Tailwinds
- **Validation:** Zod

Chapter-3

3.1 EXPECTED OUTCOME

The development and implementation of the proposed **Form Builder** system are expected to achieve the following outcomes:

- A user-friendly platform that enables both technical and non-technical users to create customized forms effortlessly using a drag-and-drop interface.
- A responsive and visually appealing interface ensuring compatibility across all devices (desktop, tablet, and mobile).
- Secure storage of form submissions with options to export data in formats such as JSON Format.
- Reduction in development time and cost for organizations that need forms, as the system eliminates the need for custom-built forms.

3.2 Future Improvement

To ensure the continued growth, relevance, and usability of the **Form Builder** system, the following enhancements and future improvements can be considered:

- **AI-Powered Form Suggestions:** Use AI to recommend field types, layouts, or validations based on user inputs or form use cases.
- **Form Optimization:** Introduce tools to analyse form performance and suggest improvements for increasing completion rates.
- **Smart Conditional Logic:** Enhance conditional logic with advanced rules, including multi-condition branching and dynamic field updates.
- **Real-Time Data Visualization:** Implement interactive dashboards for live data monitoring.
- **Custom Templates:** Enable users to save their own templates for reuse.
- **Localization and Multi-Language Support:** Improve multi-language functionality for both form creation and user interaction.
- **Offline Mode:** Allow users to create forms and collect data offline, syncing when connected to the internet.

3.3 Conclusion

Form Builder is developed to provide a web-based application that helps in the creation, customization, and management of forms in an easy manner. This tool provides users with an interface that is easily navigable, where they can create forms interactively and has the functionality to serve most of their diverse use cases.

It allows dragging and dropping of components to assemble a form and assures real-time feedback along with smooth options for customization. With the Form Builder, users can preview forms, set up validation rules, and apply conditional logic to make form submissions efficient.

The system allows for the viewing and managing of submissions efficiently with options to export data or integrate with third-party platforms.

3.4 REFERENCES

For Research: -

- [JotForm](#)
- [zapier](#)
- Chatgpt
- Taildwinds Components